**Data Analytics Capstone**

# Technical Declines in India's Unified Payments Interface: Effects of Volume Shifts and Regulatory Interventions

Final Report

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## GitHub Repository

The data files, code, database, and outputs for this report are in the public repository https://github.com/karthick2613-git/ms_upi_capstone. The folder *data/raw* holds the 33 original National Payments Corporation of India (NPCI) files, *data/processed* the cleaned panel and audit files, *data/upi.db* the SQLite database, *code* the scripts (*final_analysis.py*, *rq2_its.py*, *benchmark.py*, *bootstrap.py*, *sample_size.py*, *db_setup.py*, *workflow_figure.py*), *outputs* the figures and the illustrative watch list, and *docs* the reports. The README gives the order in which to run the scripts. Appendix A shows code excerpts and Appendix B the database schema and tuning results.

Format: this report follows the Publication Manual of the American Psychological Association, 7th edition (APA 7): Times New Roman 12-point text, double spacing, 1-inch margins, and page numbers at the top right of every page. Page counts for the text exclude the title page, tables, figures, references, and appendices.

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## Abstract

Unified Payments Interface (UPI) transactions that fail for system reasons, called technical declines (TD), cost users time and money and point to capacity or control gaps, yet public bank-level evidence on what drives them is scarce. This study estimates how sudden volume shifts, calendar peaks, and the August 2025 interface controls of the National Payments Corporation of India (NPCI) relate to the monthly TD rate of remitter banks, and tests whether machine learning (ML) and deep learning (DL) forecast next-month TD better than simple rules. The approach pairs four explanatory designs (fractional logit with bank and month effects, interrupted time series, variance components, and survival analysis) with a prediction benchmark of two naive baselines, a fractional logit, extreme gradient boosting (XGBoost), and a feed-forward neural network, all validated by rolling-origin splits in time order. The data are NPCI's public Top 50 Member Performance tables, from January 2024 to August 2026: 1,555 bank-month observations for 56 banks, held in an SQLite database, against the 1,522 required by the sample-size analysis. All work ran in Python on a two-core central processing unit (CPU), with no graphics processing unit (GPU). Volume growth showed no detectable effect on TD (a 10% jump: +0.03 percentage points [pp], 95% confidence interval [CI] −0.08 to +0.14), calendar peaks added nothing (−0.02 pp), and the August 2025 estimate changed sign across specifications, so the controls cannot be credited. TD is bank-specific (intraclass correlation 0.56) and short-lived (median spike: one month). XGBoost was the best predictor on a 653 bank-month test period, lowering root mean squared error (RMSE) from 0.958 to 0.814 pp (15%) against last month's value, with the neural network (0.831) and the fractional logit (0.823) close behind. Spike detection stayed weak (area under the precision–recall curve 0.13, base rate 0.03). The results support a monthly watch list that ranks banks by predicted TD, with capacity reviews aimed at small regional and payments banks. NPCI and member banks could run it from public tables as a refreshed dashboard.

# Introduction

## Background and Context

The Unified Payments Interface (UPI) is the real-time retail payment system run by the National Payments Corporation of India (NPCI). It lets customers of different banks send money instantly through third-party applications. Scholarly work on UPI has mostly explained why it grew (Balakrishnan, 2023; Gopika et al., 2026). The growth is large: in NPCI's member tables (NPCI, n.d.), the attempted volume of the bank members in the top 50 rose from 12.5 billion transactions in January 2024 to 26.5 billion in August 2026, so the system now handles about 2.1 times the load it did when the study window opened.

NPCI sorts every failed transaction into a business decline (BD) or a technical decline (TD). A BD has a customer-side cause, such as a wrong PIN or insufficient balance. A TD has a system-side cause, such as a connectivity, switch, or bank-processing fault, and it is the failure that banks and NPCI can act on. NPCI reports the approved, BD, and TD shares of attempted volume separately for each member bank each month, and the three sum to 100. Across all bank-months in this study the pooled TD rate is 0.41%, but it is uneven: it is 0.18% for private banks and 2.68% for regional rural banks.

Two forces make TD an operational question. The first is load, since an abrupt rise in volume can strain a bank's processing capacity. The second is regulation. An April 12, 2025 outage was linked to a surge in Check Transaction Status calls to the application programming interface (API) (Tyagi, 2025), and NPCI then limited balance checks, linked-account views, and status checks, and moved AutoPay mandate runs to non-peak hours, from August 1, 2025 (Khanduri & Mittal, 2025). NPCI sets these rules and remitter banks provision the capacity, so both are stakeholders in the answer. Central-bank incident data from Australia show that a few institutions cause many payment failures (Griffiths & Joyce, 2024), but no published bank-level evidence shows whether abrupt UPI growth or the 2025 controls moved TD in India.

## Problem Statement

Capacity decisions and API rules for UPI are made without bank-level evidence on what drives technical declines. This research aims to estimate how sudden volume shifts and regulatory interventions affect technical decline rates in UPI, in order to help NPCI and banks decide where to add capacity and which API controls to enforce. In template form: the objective of this study is to estimate $Y$ = the technical decline share of attempted transactions ($TD_{it}$, in percent) for bank $i$ in month $t$ (the unit of analysis is the bank-month), using $X$ = month-over-month log volume growth, a new-volume-peak indicator, a calendar-peak indicator (March, October, November), and a post-August-2025 indicator, over January 2024 to August 2026 in India. Success will be evaluated using average marginal effects (AME) in percentage points (pp) with 95% confidence intervals (CI) for the explanatory questions, the intraclass correlation (ICC) and spike duration for persistence, and, for prediction, root mean squared error (RMSE), mean absolute error (MAE), the area under the receiver operating characteristic curve (ROC-AUC), and the area under the precision–recall curve (PR-AUC) on a time-ordered test set.

## Purpose of the Study

The study is mainly explanatory: it estimates the size and direction of the links between TD and volume shocks, calendar peaks, and the timing of the 2025 controls, without claiming proof of cause. A second, predictive purpose is to find out whether machine learning (ML) and deep learning (DL) models, here extreme gradient boosting (XGBoost) and a neural network, forecast next-month TD and TD spikes better than naive rules and a transparent regression, because a forecast is only useful to a bank if it beats the number it already has.

## Research Problems and Research Questions

**Research Question 1 (RQ1).** Do sudden volume shifts, measured by month-over-month log volume growth and by volume exceeding a bank's previous maximum, raise a bank's TD rate?

**Research Question 2 (RQ2).** Did system-level TD change after NPCI's API controls took effect on August 1, 2025?

**Research Question 3 (RQ3).** Do predictable calendar peaks (the March fiscal year-end and the October–November festive months) raise TD, and do high-volume banks suffer more?

**Research Question 4 (RQ4).** How persistent are TD spikes, and how much of the variation in TD belongs to the bank itself?

### Contributions and Expected Value

*Practical contribution.* The study gives the first bank-level, public-data picture of UPI technical declines across 32 months, and a ranked monthly watch list that shows where failures concentrate. *Technical contribution.* It combines fractional-response regression, interrupted time series with placebo breaks, variance-components and survival models, and a rolling-origin comparison of naive, statistical, tree-based, and neural-network predictors, with every step reproducible from a public database. *Value to stakeholders.* NPCI can use the estimates to judge whether further API controls are warranted, and banks can use the watch list and the persistence results to decide whether a failure needs a capacity response or only monitoring.

## Literature Review

### Literature Review Approach

Sources came from Google Scholar, Research Papers in Economics (RePEc/IDEAS), the Munich Personal RePEc Archive (MPRA), and publisher and regulator sites, using terms such as UPI, technical decline, payment failure, outage, interrupted time series, fractional response, tabular deep learning, and language-model agents. A source was kept if it was peer reviewed, a regulator publication, an archived working paper, or attributed commentary on a UPI event, and if it bore on the UPI setting, payment reliability, a method used here, or the DL and agentic components. Table 1 lists 21 sources against the research questions (RQs), and the full list is in the Bibliography. Searches for studies of how volume shifts and the 2025 controls affected bank-level technical declines found none, so the gap is worded as “to our knowledge.”

## Summary of Key Literature

## Table 1

*Literature Relevance Matrix*

| Author (Year) | Domain/ Context | Dataset/ Setting | Method(s) | Key Findings | How It Supports RQ(s) and Project Decisions |
|---|---|---|---|---|---|
| Balakrishnan (2023) | UPI growth | India since 2010; Brazil's PIX for comparison | Descriptive, comparative | UPI is the main driver of payment growth; the paper also discusses its challenges | Context for volume as exposure; frames scale risks (RQ1) |
| Gopika et al. (2026) | UPI growth drivers | Monthly NPCI volume, July 2016–May 2022 | Autoregressive distributed lag, error correction | Internet penetration, app count, and a COVID-19 break shape volume | Volume is the outcome there and the exposure here; shows NPCI data support time series and breaks (RQ2) |
| Chawla et al. (2021) | Banking regulation, payment failure | Simulated bank cost model | Theoretical profit model | Bank profit depends on how funds are invested; stricter penalties can cause losses | Closest topic but not empirical; leaves RQ1 and RQ2 open |
| Griffiths & Joyce (2024) | Payment reliability, Australia | Reserve Bank of Australia incident data, 2021–2024 | Descriptive monitoring | Services averaged at least 99.8% availability; third-party faults led; few institutions caused many incidents | Supports a bank-heterogeneity lens (RQ4) and availability-type metrics |
| Tyagi (2025) | UPI outage commentary | April 2025 outage, India | Qualitative policy analysis | Ties the outage to a surge in API calls; argues for resilience | Motivates RQ2 and defines the intervention context |
| Khanduri & Mittal (2025) | UPI API rules | NPCI guidelines effective August 1, 2025 | Regulatory summary | Lists caps on balance and status checks and AutoPay timing | Sets the intervention date and control list (RQ2) |
| Bernal et al. (2017); Wagner et al. (2002) | Policy evaluation methods | Health-policy examples | Interrupted time series, segmented regression | Level and slope changes are estimable; autocorrelation and secular trends must be handled | Basis of the RQ2 design and its trend specifications |
| Abadie et al. (2010) | Comparative case studies | A state tobacco program | Synthetic control with placebo tests | Placebo runs give a reference distribution for an estimate | Motivates the placebo-in-time test (RQ2) |
| Papke & Wooldridge (1996) | Econometrics | 401(k) participation rates | Fractional logit (quasi-likelihood) | Bounded proportions can be modeled without impossible fitted values | Model for RQ1 and RQ3 |
| Cameron et al. (2008) | Econometrics | Simulations with few clusters | Bootstrap for clustered errors | Clustered tests over-reject with few clusters; bootstrap corrects | Inference check for about 56 bank clusters (RQ1, RQ3) |
| Kaplan & Meier (1958); Cox (1972) | Survival analysis | Incompletely observed durations | Product-limit estimator; proportional hazards | Durations and covariate effects can be estimated with censoring | Spike duration and recovery (RQ4) |
| Newey & West (1987) | Time-series inference | Autocorrelated errors | HAC covariance matrix | Standard errors stay valid under autocorrelation and heteroskedasticity | Standard errors in the RQ2 series |

| Author (Year) | Domain/ Context | Dataset/ Setting | Method(s) | Key Findings | How It Supports RQ(s) and Project Decisions |
|---|---|---|---|---|---|
| Chen & Guestrin (2016) | Machine learning | Large-scale benchmarks | Gradient-boosted trees (XGBoost) | Scalable boosting with strong benchmark results | Tree-based ML model in the benchmark |
| Shwartz-Ziv & Armon (2021) | Deep learning on tabular data | Datasets from recent deep tabular papers | XGBoost versus deep models and ensembles | XGBoost beat the deep models on every dataset and needed less tuning; an ensemble did best | Justifies a small neural network as a DL comparator, not as the default |
| Grinsztajn et al. (2022) | Deep learning on tabular data | Benchmark of medium-sized tables (about 10,000 rows) | Tree models versus deep networks | Tree-based models still beat deep learning on such data, even ignoring speed | Sets expectations for the network at 1,555 rows |
| Saito & Rehmsmeier (2015); Bergmeir & Benítez (2012) | Evaluation methods | Imbalanced classification; time-series predictors | Precision–recall analysis; order-respecting validation | PR curves inform more than ROC when positives are rare; validation must respect time | Metrics and splits for the benchmark |
| Wang et al. (2024) | Large language model (LLM) agents | Survey of LLM-based autonomous agents | Review of agent design (profile, memory, planning, action) | Describes how LLMs are built into agents that plan and act with tools | Frames the proposed alert-drafting agent (Implementation) |
| Yao et al. (2023) | LLM agents | Question-answering and interactive decision tasks | ReAct: interleaved reasoning and tool actions | Combining reasoning traces with tool use improved task performance and traceability | Supports a tool-using, auditable agent design (Implementation) |

*Note.* RQ = research question; PIX = Brazil's instant payment system; HAC = heteroskedasticity and autocorrelation consistent; ROC = receiver operating characteristic; PR = precision–recall.

The literature falls into five themes. The first is growth. Balakrishnan (2023) follows India's digital-payments expansion and compares UPI with Brazil's PIX, and Gopika et al. (2026) estimate the drivers of UPI volume from monthly NPCI data. Both treat volume as the outcome and neither studies service quality, but they show that NPCI's monthly tables support time-series work.

The second theme is reliability. Griffiths and Joyce (2024) report availability by service from regulator-collected incidents, find that third-party causes dominate, and note that a few institutions cause many incidents; this supports a bank-by-bank view and the persistence question in RQ4. Chawla et al. (2021) are closest in topic, but their profit model uses simulated

inputs. Tyagi (2025) and Khanduri and Mittal (2025) describe the April 2025 outage and the controls that followed (RQ2).

The third theme is quasi-experimental and panel methodology. Interrupted time series suits a policy that starts on a known date (Bernal et al., 2017; Wagner et al., 2002), but with one system and one date it cannot rule out coincident shocks, so this study adds placebo breaks in the spirit of Abadie et al. (2010) and fixes its decision rule in advance. Fractional logit fits TD because it is a proportion with many values near zero (Papke & Wooldridge, 1996), Newey–West standard errors protect the system-level series (Newey & West, 1987), and with about 56 bank clusters a bootstrap check is prudent (Cameron et al., 2008).

The fourth theme is prediction, including deep learning. Gradient-boosted trees (Chen & Guestrin, 2016) capture nonlinearities but give no effect estimates. Two benchmark studies find that tree ensembles still match or beat deep networks on tabular data of this kind, and that deep models are harder to tune (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2021). So here a small neural network competes against XGBoost and simple baselines under identical splits instead of being assumed superior. Because spikes are rare, precision–recall measures are more informative than accuracy or ROC curves (Saito & Rehmsmeier, 2015), and because the data are ordered in time, validation must respect order (Bergmeir & Benítez, 2012).

The fifth theme is language-model agents. Wang et al. (2024) survey agents built on large language models, which combine profile, memory, planning, and action modules, and Yao et al. (2023) show that interleaving reasoning with tool calls improves task performance and leaves a readable trace. These works inform the proposed deployment, in which an agent drafts a bank-specific note for a human analyst while all numbers come from the tested pipeline. The gap is plain: the UPI literature explains volume but not failure, the failure literature is non-Indian or theoretical, and these methods have not been applied to NPCI’s bank-level tables.

# Materials and Method

## Data Source and Acquisition

The data are NPCI's public "Top 50 Member Performance" tables for the remitter side, published at https://www.npci.org.in/product/ecosystem-statistics/upi (NPCI, n.d.). They are real, aggregate, public statistics, not synthetic data. NPCI posts one table per month from January 2024; the analysis uses 32 months, to August 2026, from 33 downloaded Excel files, one a duplicate. NPCI does not permit automated access, so no web scraping was used: each file was downloaded by hand (portal retrieved October 5, 2026), and a documented script combined them. Each row is one member in one month and carries attempted volume, the approved, BD, and TD shares, and two debit-reversal fields. The combined panel has 1,598 rows, of which 1,555 are for 56 banks and 43 are for card product lines and payment firms. After one rename mapping the panel holds 62 entities. Table 2 defines every field.

**Table 2**

*Data Dictionary*

| Variable | Definition | Type | Allowed values/range | Missing values | Notes |
|---|---|---|---|---|---|
| month | Reporting month | Date | 2024-01 to 2026-08 | None | From sheet title; one file by file name |
| rank | NPCI rank by volume in the month | Integer | 1–50 | None | Descriptive only |
| bank_raw, bank | Name as published; harmonized name | Text | 159 raw spellings; 63 names | None | Rule-based map |
| entity | Bank after merger/rename mapping | Text | 62 entities (56 banks) | None | Cluster and fixed-effect unit |
| volume_mn | Attempted remitter transactions (million) | Numeric | Observed 14.1–6,846 | None | Exposure and weight |
| approved_pct | Share of attempts approved (%) | Numeric | 0–100 (54.1–97.2) | None | Approved + BD + TD = 100 |
| bd_pct | Business decline share (%) | Numeric | 0–100 (2.2–27.0) | None | Placebo outcome (RQ2) |
| td_pct | Technical decline share (%) | Numeric | 0–100 (0.0–36.3) | None; 70 zeros kept | Dependent variable (Y) |
| debit_reversal_mn | Debit reversals (million) | Numeric | 0–23.2 (observed) | 29 missing (1.8%), left blank | Not used in models |
| debit_reversal_success_pct | Reversals completed (%) | Numeric | 0–100 | 3 missing (0.2%), left blank | Not used in models |

| Variable | Definition | Type | Allowed values/range | Missing values | Notes |
|---|---|---|---|---|---|
| is_bank | False for card lines and non-bank firms | Logical | True/False | None | Filter (43 rows False) |
| source_file, month_source | File and how the month was set | Text | 33 files | None | Audit trail |
| lnV, ΔlnV | Log volume; month-over-month log growth | Numeric | ΔlnV −1.10 to 0.81 | ΔlnV missing for each bank's first month | X (RQ1) |
| NewPeak | 1 if volume exceeds bank's earlier maximum | Binary | 0/1 | Undefined in first month | X (RQ1) |
| Post, Peak | Post: 1 from 2025-08. Peak: March, October, November | Binary | 0/1 | None | X (RQ2, RQ3) |
| Spike | 1 if robust z of TD > 3 and TD ≥ 1.0% | Binary | 0/1 | No spike if median absolute deviation is 0 | Outcome (RQ4, classifier) |

*Note.* Source: NPCI (n.d.). BD = business decline; TD = technical decline; pct = percent; mn = million.

## Inclusion and Exclusion Criteria

Included are remitter-side rows for members of NPCI's top-50 list in a given month, for the 56 entities that are banks, from January 2024 to August 2026. Excluded from bank-level models, but kept and flagged in the database, are the 43 rows for credit-card product lines and non-bank payment firms, because their business model and failure profile differ. Also excluded are the duplicate file, beneficiary-side tables (not collected), and months after August 2026. Growth variables are undefined for a bank's first month or after a gap in the top-50 list, and those rows drop out of the models that use them.

## Exploratory Data Analysis

Exploratory data analysis (EDA) used univariate, bivariate, and multivariate views, and its findings fixed the modeling choices. Table 3 shows that TD is highly skewed: the median is 0.33%, the mean 0.95%, and the maximum 36.31%, so a few observations carry much of the variation and a model on the raw scale would be dominated by them. This is the first reason for using a fractional logit. Volume is skewed too, so it enters models in logs.

**Table 3**

*Descriptive Statistics, Bank-Months (n = 1,555)*

| Variable | Mean | SD | Minimum | Median | 99th pct. | Maximum |
|---|---|---|---|---|---|---|
| Attempted volume (million) | 392.1 | 776.5 | 14.1 | 103.4 | 4,935.1 | 6,845.8 |

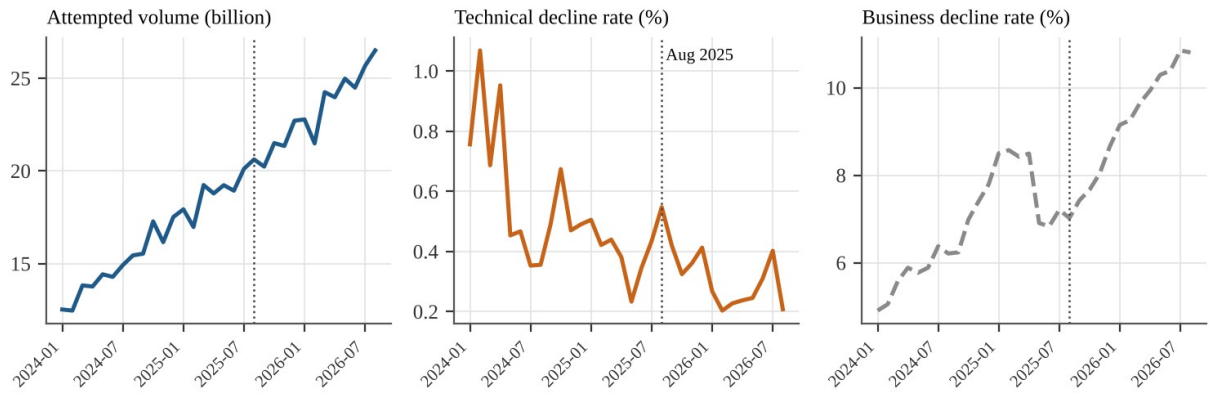
| Variable | Mean | SD | Minimum | Median | 99th pct. | Maximum |
|---|---|---|---|---|---|---|
| Approved share (%) | 90.51 | 4.39 | 54.14 | 91.80 | 96.38 | 97.18 |
| BD share (%) | 8.55 | 3.71 | 2.23 | 7.59 | 22.78 | 26.97 |
| TD share (%) | 0.95 | 1.91 | 0.00 | 0.33 | 10.11 | 36.31 |

*Note.* SD = standard deviation; pct. = percentile. TD varies between banks (SD 1.27 pp) and within banks (SD 1.51 pp).

Figure 1 gives the system view. Attempted volume of the bank members roughly doubled over the 32 months (about 2.4% growth a month), while the volume-weighted TD rate fell from 0.76% to 0.21%, peaking at 1.07% in February 2024. The correlation of monthly volume with TD is −0.71, but both series trend, so it does not show that volume lowers TD. System growth did not coincide with worse reliability, so any volume effect has to be sought in bank-level shocks.

**Figure 1**

*System-Level Trends in Volume, Technical Decline Rate, and Business Decline Rate*



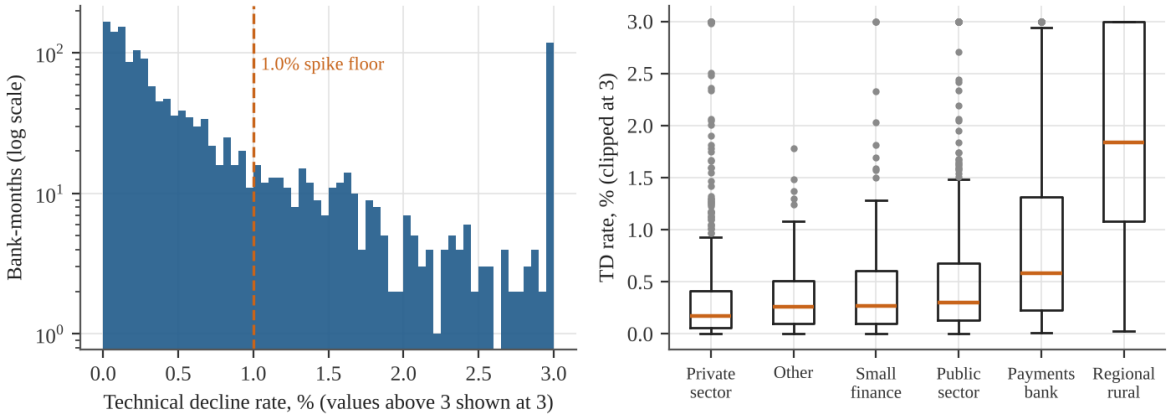

*Note.* Volume-weighted monthly values for bank members. The dotted line marks August 2025. Source: author's calculation from NPCI (n.d.).

Figure 2 and Table 4 show that TD is concentrated by bank type. Regional rural banks have a median bank-month TD of 1.84% against 0.17% for private banks and 0.30% for public sector banks, although they carry only 1.6% of volume. Payments banks have the highest spike rate (11.3% of bank-months). Public sector banks carry 61% of volume, so a system average mostly describes them and would miss the small banks where TD is concentrated. Figure 3 adds

that larger banks have lower TD (Spearman correlation −0.51 across 56 banks), so cross-bank comparisons are confounded by size and the models use bank fixed effects.

**Figure 2**

*Distribution of Technical Decline Rates and Differences by Bank Group*




*Note.* Left: histogram of bank-month TD (log count axis; values above 3% shown at 3). Right: boxplots by group, ordered by median. Groups are assigned by rule from bank names. Source: author's calculation from NPCI (n.d.).

**Table 4**

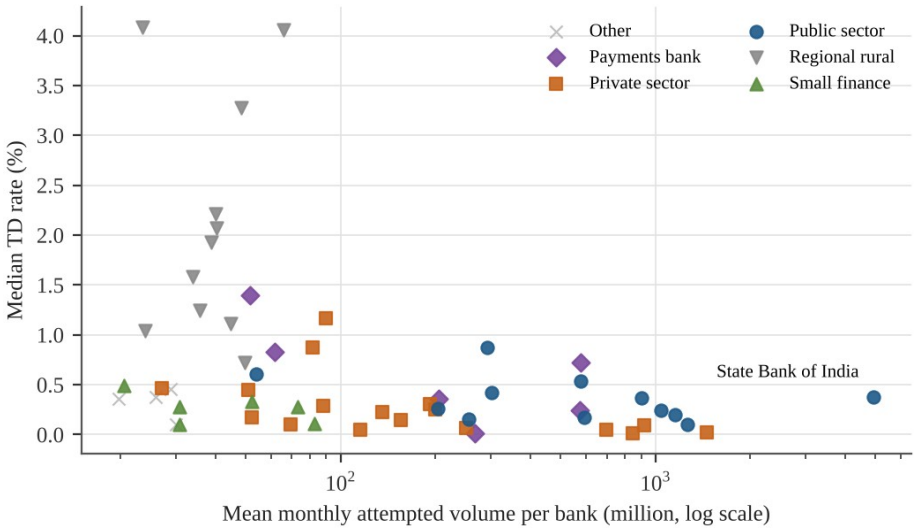
*Technical Decline Rate and Spike Frequency by Bank Group*

| Group | Banks | Bank-months | Mean TD (%) | Median TD (%) | Pooled TD (%) | Spike rate (%) | Share of volume (%) |
|---|---|---|---|---|---|---|---|
| Public sector | 12 | 384 | 0.61 | 0.30 | 0.43 | 7.6 | 60.7 |
| Private sector | 17 | 537 | 0.43 | 0.17 | 0.18 | 3.9 | 28.4 |
| Small finance | 6 | 143 | 0.50 | 0.27 | 0.45 | 8.4 | 1.1 |
| Payments bank | 6 | 151 | 1.20 | 0.58 | 0.69 | 11.3 | 7.8 |
| Regional rural | 11 | 251 | 2.88 | 1.84 | 2.68 | 8.8 | 1.6 |
| Other | 4 | 89 | 0.36 | 0.26 | 0.36 | 3.4 | 0.4 |

*Note.* Mean and median are over bank-month rates; pooled TD is total technical declines divided by total attempts. Spike: robust z-score of TD above 3 and TD at least 1.0%. "Other" combines foreign banks and one co-operative bank.

**Figure 3**

*Median Technical Decline Rate Against Average Volume, by Bank*

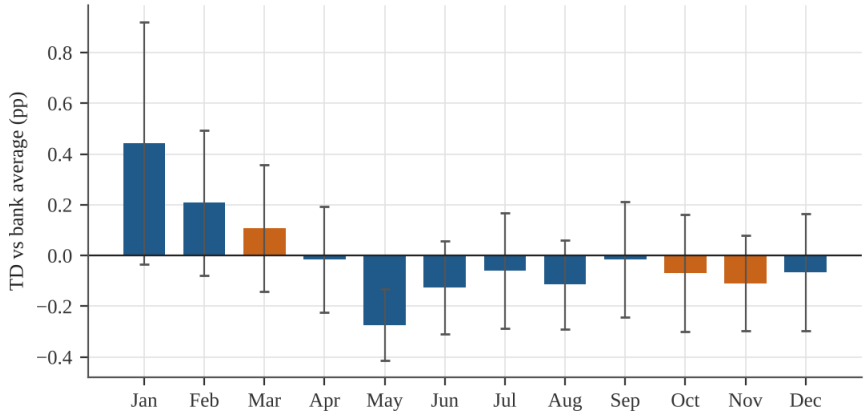



*Note.* Each point is one bank entity (n = 56); the horizontal axis is on a log scale. Source: author's calculation from NPCI (n.d.).

Figure 4 tests the calendar idea behind RQ3. After removing each bank's average, TD is highest in January (+0.44 pp) and February (+0.21 pp), slightly above average in March (+0.11 pp), and not raised in October (−0.07 pp) or November (−0.11 pp). The early-year values reflect early 2024, when 47 of the 104 spike-months occurred (Figure 6). Figure 5 relates volume growth to the change in TD for the same bank. The points center on zero and the rank correlation is 0.003 ($p$ = .91, $n$ = 1,489), so ordinary monthly growth, whose SD is only 0.075, does not visibly move TD. The tails differ: in 14 bank-months with growth above 0.20, TD rose by 1.06 pp on average against −0.07 pp for the rest, driven by three small banks. Three cases are anecdotal, but they suggest where an effect would appear: rare, large jumps at small banks, which an average-effect model has little power to detect.

**Figure 4**

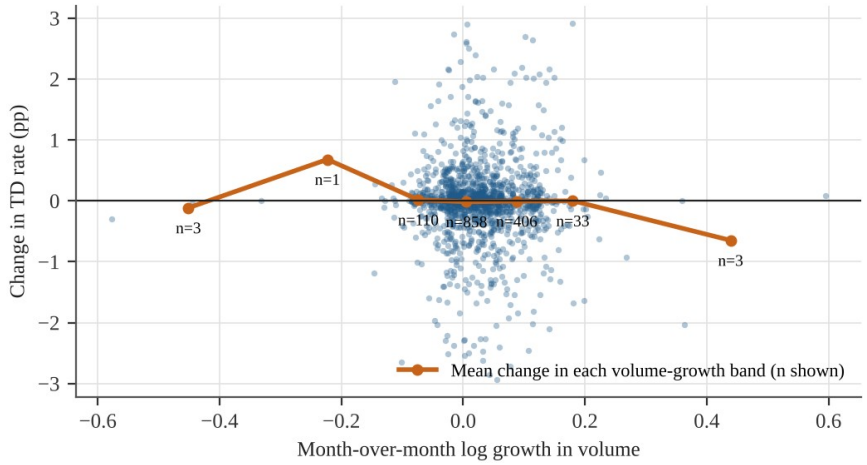
*Technical Decline Rate by Calendar Month, Relative to Each Bank's Average*




*Note.* Bars show the mean deviation in pp with 95% intervals; orange bars mark the hypothesized peak months. Source: author's calculation from NPCI (n.d.).

**Figure 5**

*Monthly Volume Growth and the Change in Technical Decline Rate, Same Bank*



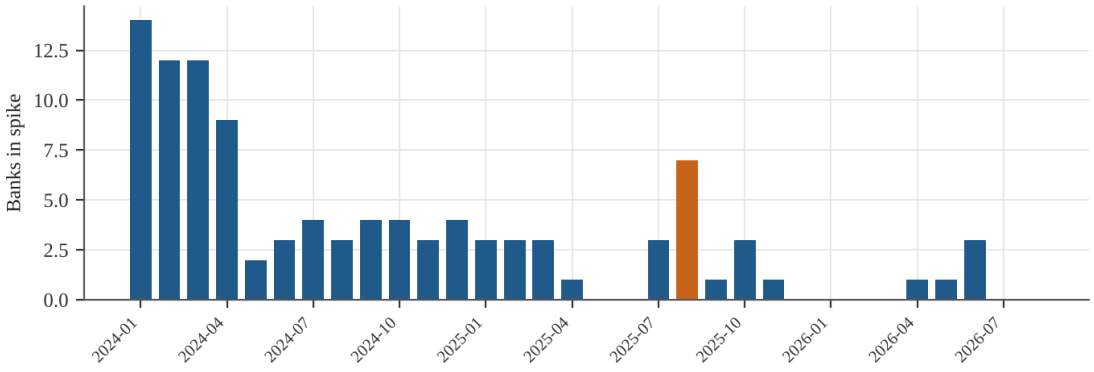

*Note.* Points are bank-months (|ΔTD| < 3 pp and |ΔlnV| < 0.8 shown); the line gives the mean change within growth bands, with counts. Source: author's calculation from NPCI (n.d.).

Figure 6 shows when spikes occur: 14, 12, 12, and 9 banks were in spike in January to April 2024, they thin out through 2025, and they nearly disappear in early 2026. August 2025, when the controls began, has a small cluster of 7 banks, smaller than the 2024 cluster. TD was therefore already falling when the controls arrived, so a simple before-and-after comparison for RQ2 confounds the intervention with an existing trend. The multivariate view in Figure 7 agrees:

TD relates negatively to log volume (Spearman −0.41) and the approved share (−0.43) and only weakly to BD (0.19) and volume growth (0.06), which points to bank size and level, not growth, as the structure to model.

**Figure 6**

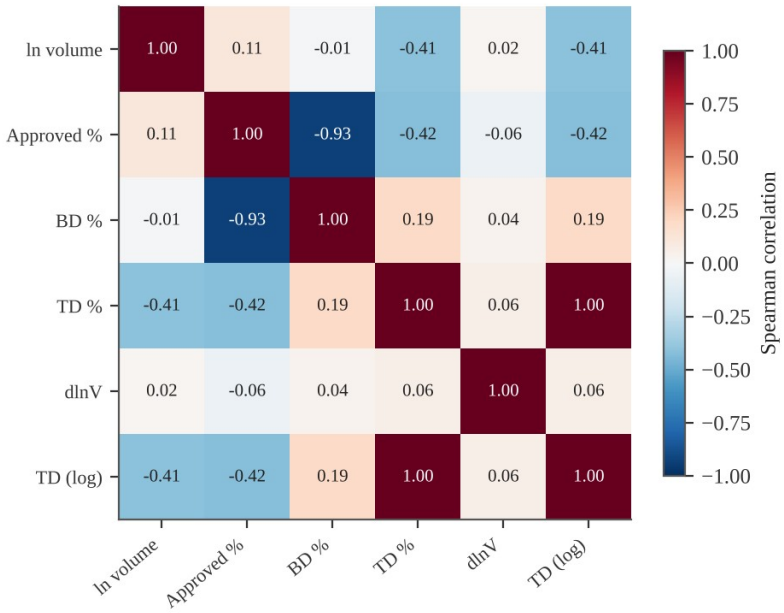
*Number of Banks in a Technical Decline Spike, by Month*




*Note.* A bank is in a spike if its robust z-score of TD exceeds 3 and TD is at least 1.0%. The orange bar marks August 2025. Source: author's calculation from NPCI (n.d.).

**Figure 7**

*Spearman Correlation Heatmap of Core Variables (Bank-Months)*




*Note.* ln volume = natural log of attempted volume; ΔlnV = monthly log growth; TD (log) = logit of the TD share. Source: author's calculation from NPCI (n.d.).

## Research Hypotheses

Table 5 states the null ($H_0$) and alternative ($H_a$) hypotheses for each research question. Every test uses a 5% significance level ($\alpha = .05$, two-sided), and the results section reports the decision for each.

**Table 5**

*Hypotheses and Statistical Methods by Research Question*

| RQ | $H_0$ (null) | $H_a$ (alternative) | Method and test |
|---|---|---|---|
| RQ1 | $H_{01a}$: $\beta(\Delta \ln V) = 0$<br>$H_{01}$b: $\beta(\text{NewPeak}) = 0$ | $H_{a1a}$: $\beta(\Delta \ln V) \neq 0$<br>$H_{a1}$b: $\beta(\text{NewPeak}) \neq 0$ | Fractional logit, bank and month fixed effects, bank-clustered Wald test; bank bootstrap check |
| RQ2 | $H_{02}$: level shift $\delta = 0$ at August 2025 | $H_{a2}$: $\delta \neq 0$ | Interrupted time series, five specifications, Newey–West *t* test; placebo breaks |
| RQ3 | $H_{03a}$: $\beta(\text{Peak}) = 0$<br>$H_{03}$b: $\beta(\text{Peak} \times \text{exposure}) = 0$ | $H_{a3a}$: $\beta(\text{Peak}) \neq 0$<br>$H_{a3}$b: $\beta(\text{Peak} \times \text{exposure}) \neq 0$ | Fractional logit with bank effects, trend, and Post; clustered Wald test; bank bootstrap |
| RQ4 | $H_{04a}$: ICC = 0<br>$H_{04}$b: AR(1) $\rho = 0$<br>$H_{04}$c: hazard ratios = 1 | $H_{a4a}$: ICC > 0<br>$H_{a4}$b: $\rho \neq 0$<br>$H_{a4}$c: hazard ratios ≠ 1 | Random-intercept model; clustered AR(1); Kaplan–Meier and Cox model |

*Note.* ICC = intraclass correlation coefficient; AR(1) = first-order autoregression; NewPeak = volume above the bank's earlier maximum; exposure = bank's mean log volume (standardized).

## Minimum Sample Size

Sample size was set for each question by a power or confidence-interval formula, with $\alpha = .05$ (two-sided, $z = 1.960$) and power 0.80 ($z = 0.842$), so $K = 1.960 + 0.842 = 2.802$. For RQ1, $n = (K\sigma_e / (\beta\ SD_x))^2$, where $\sigma_e = 1.0508$ is the residual SD of the logit of TD in a model with bank and month effects, $SD_x = 0.1258$ is the SD of log volume left after those effects, and the smallest effect of interest, $\beta = 0.60$ logit units, is a planning choice that equals a 5.9% rise in the odds of a technical decline for a 10% volume increase. Table 6 shows the calculations: RQ2 uses a confidence-interval half-width, RQ3 Fisher's *z*, RQ4a a confidence interval for the ICC, and RQ4b the Schoenfeld formula for events.

**Table 6**

*Required and Available Sample Size by Research Question*

| RQ | Method and formula | Calculation | Required | Available | Verdict |
|---|---|---|---|---|---|
| RQ1 (level) | Slope power: $n = (K\sigma_e/(\beta\, SD_x))^2$ | $(2.802 \times 1.0508 / (0.60 \times 0.1258))^2 = 1{,}521.8$ | 1,522 bank-months | 1,555 | Met |
| RQ1 (growth) | Same, SD of ΔlnV | $SD_x = 0.07536$: $(2.802 \times 1.0508 / (0.60 \times 0.07536))^2 = 4{,}239$ | 4,239 bank-months | 1,489 | Not met at $\beta = 0.60$; detectable $\beta = 1.01$ |
| RQ2 | CI half-width 1.96 × SE, SE inflated by $\sqrt{((1+\rho)/(1-\rho))}$ | $\sigma = 0.1347$ pp; $\rho = 0.224$; inflation 1.255; SE = 0.116 at 32 months; target 0.25 pp | 26 months | 32 months | Met |
| RQ3 | Fisher's z: $n = (K/\text{atanh } r)^2 + 3$ | $r = 0.20$: $(2.802/0.2027)^2 + 3 = 194$ peak bank-months; 194/(342/1,555) = 882 | 194 peak (882 bank-months) | 342 peak | Met |
| RQ4a (ICC) | $\text{Var(ICC)} = 2(1-\rho)^2[1+(m-1)\rho]^2 / (m(m-1)(k-1))$ | $\rho = 0.557$, m = 27.8; k = 56 gives half-width 0.097; half-width 0.10 needs k = 52.3, rounded to 53; 53 × 27.8 = 1,473 | 53 banks (1,473 bank-months) | 56 banks | Met |
| RQ4b (duration) | Schoenfeld: $d = K^2/(p(1-p)(\ln HR)^2)$ | HR = 2, p = 0.5: 7.85/(0.25 × 0.4805) = 65.3, so 66 episodes; 66/(69/1,555) = 1,487; EPV check 3 × 10 = 30 | 66 episodes (1,487 bank-months) | 69 episodes | Met |

*Note.* HR = hazard ratio; EPV = events per variable; m = months per bank; k = banks. Bank-month equivalents use observed rates (342 peak bank-months and 69 episodes per 1,555 bank-months). Computed in code/sample_size.py.

Expressed in bank-months, the requirements are 1,522, 882, 1,473, and 1,487, so the binding sample size is $N = \max(1{,}522; 882; 1{,}473; 1{,}487) = 1{,}522$, set by RQ1 and met by the 1,555 bank-months collected, with a margin of 33. The RQ2 series needs 26 months and has 32, and the smallest level shift it can detect with 80% power is 2.802 × 0.116 = 0.325 pp. Because growth has a much smaller SD than the volume level, the growth version of RQ1 needs 4,239 pairs against 1,489 available. The smallest detectable effect is $\beta = 1.01$, about a 10% rise in the odds of a technical decline per 10% volume jump, so smaller effects are read as inconclusive, not as absent.

## Data Splitting Procedure

The prediction task uses information at month *t* to predict a bank's TD and spike status at month *t* + 1, which gives 1,425 usable bank-month pairs. Splitting is by target month, never at random, so no model sees later months than it predicts. Table 7 shows the three partitions. The

training partition (target months through December 2024) is where models first fit; the validation partition (January to June 2025) selects hyperparameters and the spike-alert thresholds; and the test partition (July 2025 to August 2026) is touched only for final evaluation. In the rolling-origin scheme, every month is predicted by a model refit on all earlier target months (Bergmeir & Benítez, 2012). Training accuracy is reported from a model fit once on all pairs before the first test month (772 pairs), and test accuracy from the rolling predictions.

**Table 7**

*Data Splitting by Target Month*

| Partition | Target months | Bank-month pairs | Spikes | Use |
|---|---|---|---|---|
| Training (initial) | Feb 2024–Dec 2024 | 478 | 46 | First model fit; rolling refits add later months |
| Validation | Jan 2025–Jun 2025 | 294 | 10 | Hyperparameters; alert thresholds |
| Test | Jul 2025–Aug 2026 | 653 | 20 | Final out-of-sample evaluation (spike base rate 3.1%) |

*Note.* Every test month is predicted by a model fit on earlier target months only. Scalers and class weights are fit on training rows only.

### Database Set-Up and Data Access

The cleaned panel is stored in a SQLite database (*data/upi.db*) built by *code/db_setup.py*. It has three tables: *panel* (1,598 member-months, keyed by month and entity), *entity_crosswalk* (each of the 159 raw name spellings mapped to a harmonized name and entity), and *file_log* (one row per source file). A view, *v_bank_month*, returns the 1,555 bank rows used in the models. Integrity checks run on every build: one row per entity-month, approved + BD + TD within 0.01 pp of 100, 32 months, 62 entities, and no duplicate keys. Code reads the data through Structured Query Language (SQL) or the equivalent CSV file; for example, *SELECT month, SUM(volume_mn*td_pct)/SUM(volume_mn) FROM v_bank_month GROUP BY month* returns the volume-weighted TD series that Figure 1 plots. Appendix B gives the schema and more queries, and any reviewer can rebuild the database from the public raw files.

**Statistical and Modeling Methods**

RQ1 and RQ3 use a fractional logit, a generalized linear model with a binomial family and logit link, fit to the TD share with bank fixed effects and standard errors clustered by bank (Papke & Wooldridge, 1996). RQ1 adds month fixed effects in two forms: Model A with log volume growth and the new-peak indicator, and Model B, which adds last month's TD. Effects are average marginal effects, AME = mean[$\hat{\beta} \cdot \hat{\mu}(1 - \hat{\mu})$] × 100 in pp, and a 10% jump multiplies ΔlnV by ln 1.1. RQ3 uses bank effects, a linear trend, Post, Peak, and Peak interacted with the bank's standardized mean log volume. With about 56 clusters, a bank-level pairs bootstrap (400 resamples) gives a second set of intervals (Cameron et al., 2008).

RQ2 models the volume-weighted monthly TD with a level change at August 2025 under five specifications: a step only; a linear trend plus a step (the planned model); the 12 months before the break against all months after; and the log versions of the first two. Newey–West standard errors with two lags handle autocorrelation (Newey & West, 1987). The models are re-run with the break at each of 21 candidate months with at least six months on both sides, a placebo-in-time test (Abadie et al., 2010; Bernal et al., 2017), and BD serves as a placebo outcome. The decision rule was fixed in advance: the August step is credited only if its sign agrees across all five specifications and it ranks in the top two of the 21 candidates. RQ4 uses a random-intercept model on the logit of TD to get the ICC, $\sigma^2_u/(\sigma^2_u + \sigma^2_e)$, a pooled AR(1) on bank-demeaned logit TD with a half-life of ln(0.5)/ln(ρ), and Kaplan–Meier and Cox models for the duration of spike episodes (Cox, 1972; Kaplan & Meier, 1958).

The prediction benchmark compares five regression models for next-month TD: the naive rule (next month equals this month), the bank's expanding historical mean, a fractional logit, XGBoost (Chen & Guestrin, 2016), and a feed-forward neural network (multilayer perceptron, MLP) with one or two hidden layers. A matching set of classifiers predicts next-month spikes: the naive rule (still in spike), a regularized logistic regression, an XGBoost classifier, and the MLP classifier. The two baselines stand for current practice. Model choice follows the data: the logit gives interpretable effects on a bounded proportion, XGBoost captures nonlinearity in small

tabular data, and the MLP tests whether a DL model adds anything, which the literature gives reason to doubt (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2021). A model is preferred only if it beats the baselines on the test partition, judged by a bank-level bootstrap interval (1,000 resamples) for each RMSE difference.

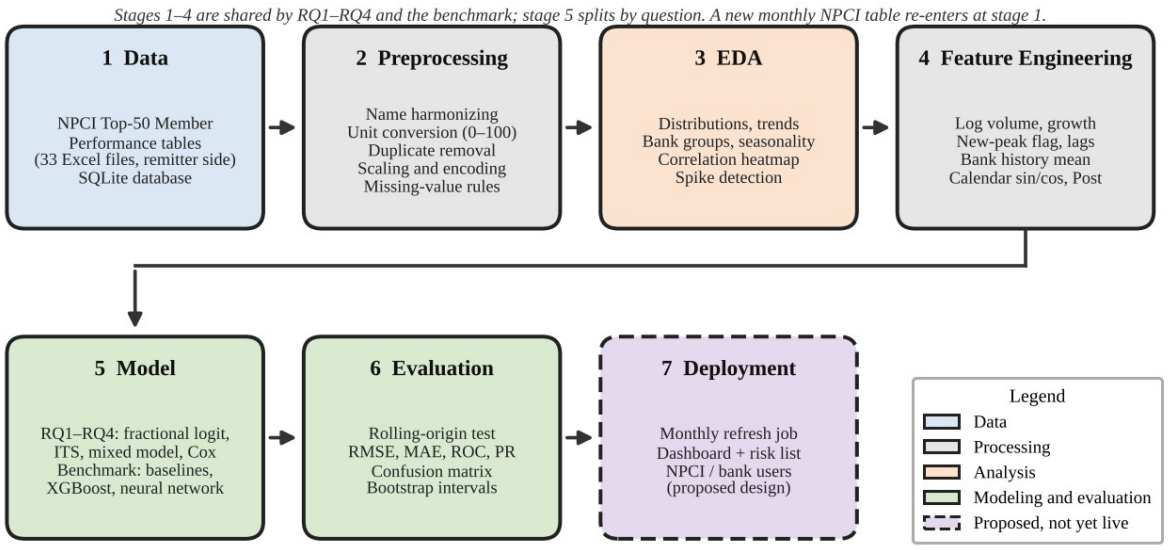
## Architecture Diagram and Workflow

### System Overview

The pipeline turns NPCI's monthly public tables into answers to RQ1 to RQ4 and into a next-month risk list. Figure 8 shows its seven stages: data, preprocessing, EDA, feature engineering, model, evaluation, and deployment. Stages 1 to 4 are shared by all questions; stage 5 branches into the explanatory models and the prediction benchmark; stage 6 scores them on held-out months; and stage 7, shown with a dashed border because it is a proposed design that is not yet live, would refresh the outputs each month when NPCI posts a new table.

**Figure 8**

*End-to-End Workflow of the Proposed System*




*Note.* Solid borders mark stages run in this study; the dashed border marks the proposed deployment stage. Colors group stages by role, as the legend shows. Source: author.

## Workflow Components

### *Data Ingestion*

The inputs are 33 Excel files downloaded from NPCI's portal (Figure 8, stage 1). A script reads each file, sets the month from the sheet title (or the file name when the title is wrong), converts all percentages to a 0 to 100 scale, and appends the rows to one table, which is then loaded into the SQLite database.

### *Data Preprocessing*

Table 8 lists each data issue and its treatment, under one rule: change a value only when the source plainly shows an error, flag judgment calls, and keep every real observation, because extreme TD months are the phenomenon under study.

**Table 8**

*Data Preparation Decisions*

| Issue | Count | Treatment | Reason |
|---|---|---|---|
| Missing values | 29 of 1,598 (1.8%) in reversal volume; 3 (0.2%) in reversal success; none in core fields | Left blank; no imputation; the two fields are not used | Imputing unused fields adds assumptions without benefit |
| Zero TD | 70 rows (4.4%) | Kept; floored at 0.0025 pp only when a logit is computed | Reporting resolution is 0.01 pp, so a zero may be a rounded small value |
| Outliers | 166 bank-months above the IQR fence (2.18%); 104 spikes in 37 banks; maximum 36.31% | Retained and flagged; training targets for XGBoost and MLP capped at 10 pp; evaluation on uncapped values | Deleting spikes would remove the signal; capping limits leverage during fitting |
| Duplicates | 1 duplicate file; 2 duplicate listings of one bank | File dropped; listings combined by volume weighting | A duplicated month would inflate persistence |
| Inconsistent formats | 6 files with fractions; others text | All converted to 0–100 numeric | Approved + BD + TD must equal 100 (verified) |
| Mislabeled month | 1 file (50 rows) titled July | Treated as August 2026 by file name; flagged for portal check | Content differs from the July file |
| Name variants | 159 raw spellings | Mapped to 63 names; one rename merged into an entity (12 rows) | One bank must be one unit |
| Non-bank members | 43 rows (2.7%) | Flagged is_bank = False; excluded from bank models | Different failure profile |
| Scaling | MLP and logistic regression inputs | Standardized to z-scores; scaler fit on training rows in each refit | Gradient-based models need comparable scales; avoids leakage |

| Issue | Count | Treatment | Reason |
|---|---|---|---|
| Encoding | Bank group; month; bank and month effects | Five 0/1 group indicators (“Other” is the reference); sin and cos of the target month; dummy variables as fixed effects | Models need numeric inputs; sin/cos keeps December near January |

*Note.* IQR = interquartile range. Counts come from code/final_analysis.py and code/db_setup.py.

Missing values are rare and confined to two unused fields, so none were imputed, and the core fields (volume, approved, BD, and TD shares) are complete in all 1,598 rows. Lagged features are undefined for a bank’s first month, and those rows are dropped from models that use them. Outliers were retained and flagged, not removed: a robust *z*-score from each bank’s own median and median absolute deviation marks a spike when it exceeds 3 and TD is at least 1.0%. For scaling and encoding, inputs to the MLP and the logistic regression are standardized with statistics from the training rows only, tree models and the fractional logit use raw values, group membership is one-hot encoded, and the calendar month is encoded by its sine and cosine.

### *Exploratory Data Analysis*

EDA (Figures 1 to 7, Tables 3 and 4) led to a fractional logit (skewed TD), bank fixed effects (size and group drive level), a trend control in RQ2 and RQ3 (TD was already falling), and a spike definition relative to each bank’s history.

### *Feature Engineering*

Table 9 lists the features. They follow the research questions: log volume and its growth for RQ1, a new-peak indicator, calendar indicators for RQ3, and the bank’s own TD history for persistence and prediction. Every feature for predicting month $t + 1$ uses information available at month $t$, and the historical mean is an expanding mean, never a full-sample mean. Spike labels use full-sample bank medians, so they are targets, not inputs.

**Table 9**

*Feature Set and Usage*

| Feature | Original/ engineered | Type | Reason for inclusion | Used in |
|---|---|---|---|---|
| $TD_t$, $TD_{t-1}$ | Original (lagged) | Numeric | Persistence of TD ($\rho = 0.32$) | All predictive models |
| $BD_t$ | Original | Numeric | User-side stress may signal load | Logit, XGBoost, MLP |

| Feature | Original/ engineered | Type | Reason for inclusion | Used in |
|---|---|---|---|---|
| $\ln V_t$ | Engineered (log) | Numeric | Size and exposure; removes skew | Logit, XGBoost, MLP |
| $\Delta \ln V_t$ | Engineered | Numeric | Sudden volume shift (RQ1) | All regression models |
| $\text{NewPeak}_t$ | Engineered | Binary | Volume above prior maximum (RQ1) | All regression models |
| Bank history mean TD | Engineered (expanding) | Numeric | Bank-specific level without leakage | Baseline and models |
| sin, cos of target month; Peak | Engineered | Numeric/ binary | Seasonality and peaks (RQ3) | Logit, XGBoost, MLP |
| Bank group (5 indicators) | Engineered (rule) | Binary | TD level differs by group | Logit, XGBoost, MLP |
| Post; bank and month effects | Engineered | Binary/ dummies | Intervention timing; remove stable differences | ITS, RQ1, RQ3 |

*Note.* ITS = interrupted time series. The predictive models use 15 features in all.

### *Model Development*

Models were built in the order of the research questions and then the benchmark (Figure 8, stage 5). Hyperparameters were chosen on the validation partition from small grids (XGBoost: depth 2 to 4, 150 or 300 trees; MLP: one layer of 16 units or two of 32 and 16, penalty 0.001 or 0.1; logistic regression: three penalty strengths) with the learning rate, subsampling, and seed fixed at 0.05, 0.8, and 42 (Appendix B). The MLP averages three random starts. The spike classifiers address class imbalance by weighting the positive class (XGBoost, logistic regression) or oversampling spikes in the training rows (MLP), and each alert threshold maximizes $F_1$ on the validation partition.

### *Model Evaluation*

Regression models are scored by $\text{RMSE} = \sqrt{((1/n)\,\Sigma(y_i - \hat{y}_i)^2)}$, $\text{MAE} = (1/n)\,\Sigma|y_i - \hat{y}_i|$, and the coefficient of determination $R^2$ on training and test partitions. Classifiers are scored by precision = TP/(TP + FP), recall = TP/(TP + FN), the confusion matrix at the validated threshold, ROC-AUC, and PR-AUC, which equals the average precision $\Sigma(R_n - R_{n-1})P_n$. Here TP, FP, and FN are true positives, false positives, and false negatives, and R and P are recall and precision.

With 20 spikes among 653 test pairs (3.1%), accuracy alone would mislead, so PR-AUC is judged against that base rate (Saito & Rehmsmeier, 2015).

### *Deployment*

Deployment is a proposed design, described under Implementation and User Benefit: a monthly job that rebuilds the database, refits the chosen models, and publishes a ranked watch list and dashboard for NPCI analysts and bank operations teams.

## Tools and Technologies

Analysis used Python 3.13 with pandas and NumPy for data handling, SciPy and statsmodels for the statistical models, scikit-learn for the neural network, logistic regression, and metrics, XGBoost for gradient boosting, matplotlib for figures, and SQLite through the standard *sqlite3* module for storage. All computing ran on a cloud virtual machine with two central processing unit (CPU) cores and no graphics processing unit (GPU); the networks are small and train quickly on 1,425 rows. The seed is 42 throughout, and the repository scripts regenerate every table and figure.

# Results

## Model Performance

Table 10 compares the five next-month TD predictors on the 653 test pairs from July 2025 to August 2026, with training error from a model fit on the 772 earlier pairs. XGBoost is the best model: it has the lowest test RMSE (0.814 pp) and MAE (0.433 pp) and the highest $R^2$ (0.411). Against the naive rule, which is the benchmark a bank already has, it lowers RMSE by 15.0% (0.958 to 0.814) and MAE by 7.1%. The fractional logit (0.823) and the neural network (0.831) are within 0.02 pp of it, and the bank-level bootstrap interval for the RMSE difference from the naive rule excludes zero for all three (XGBoost: −0.246 to −0.055 pp), whereas the historical mean does no better than the naive rule (interval −0.148 to +0.146). The three best models do not differ reliably from each other, so the choice rests on MAE, where only XGBoost clearly beats the naive rule.

**Table 10**

*Model Performance Comparison for Next-Month TD (Rolling-Origin Test, July 2025–August 2026)*

| Model | Features used | Validation method | RMSE train / test (pp) | MAE train / test (pp) | Test $R^2$ | Key observation |
|---|---|---|---|---|---|---|
| Naive (last month) | TD at t | Rolling origin, no fitting | 1.578 / 0.958 | 0.728 / 0.466 | 0.184 | Hard baseline; TD is persistent |
| Bank history mean | Expanding bank mean | Rolling origin, no fitting | 1.883 / 0.968 | 0.977 / 0.602 | 0.167 | No gain over naive |
| Fractional logit | 15 features (Table 9) | Rolling origin, refit monthly | 1.273 / 0.823 | 0.663 / 0.473 | 0.397 | Interpretable; RMSE gain of 14%; MAE no better than naive |
| **XGBoost (best)** | 15 features | Rolling origin, tuned on validation | **0.876 / 0.814** | **0.447 / 0.433** | **0.411** | **Lowest error on both metrics; 15% RMSE gain** |
| Neural network (MLP) | 15 features, standardized | Rolling origin, tuned on validation | 1.267 / 0.831 | 0.624 / 0.471 | 0.386 | Close to the logit; no gain from DL |

*Note.* Lower is better. Training error comes from a model fit once on 772 pairs before July 2025; test error from monthly refits. Training RMSE of the baselines is higher than their test RMSE because the training period holds the volatile early-2024 spikes. 95% bootstrap intervals for the RMSE difference from the naive rule: logit −0.232 to −0.046; XGBoost −0.246 to −0.055; MLP −0.220 to −0.040. XGBoost: depth 3, 150 trees; MLP: 16 hidden units, penalty 0.001.

Spike detection is much weaker, and Table 11 shows why. The test partition has 20 spikes in 653 pairs (3.1%). The XGBoost classifier ranks spikes best (ROC-AUC 0.74, PR-AUC 0.13, about 4.3 times the base rate), but at its validated threshold it raises 29 alerts, 4 of them correct (precision 0.14, recall 0.20) and misses 16 spikes. The naive rule, which flags banks already in a spike, scores a higher $F_1$ (0.20 against 0.16). The logistic regression and the neural network catch none. The XGBoost classifier also overfits (training ROC-AUC 1.00, test 0.74). No model is a reliable spike alarm; XGBoost is the best of a weak set only in how well it orders risk.

**Table 11**

*Spike Detection Performance on the Test Partition (20 Spikes in 653 Pairs)*

| Model | Alerts | TP | FP | FN | Precision | Recall | ROC-AUC | PR-AUC | Train ROC-AUC |
|---|---|---|---|---|---|---|---|---|---|
| Naive (still in spike) | 20 | 4 | 16 | 16 | 0.20 | 0.20 | 0.59 | 0.065 | n/a |
| Logistic regression | 1 | 0 | 1 | 20 | 0.00 | 0.00 | 0.69 | 0.100 | 0.82 |
| **XGBoost classifier** | 29 | 4 | 25 | 16 | 0.14 | 0.20 | **0.74** | **0.133** | 1.00 |
| Neural network (MLP) | 8 | 0 | 8 | 20 | 0.00 | 0.00 | 0.56 | 0.040 | 0.93 |

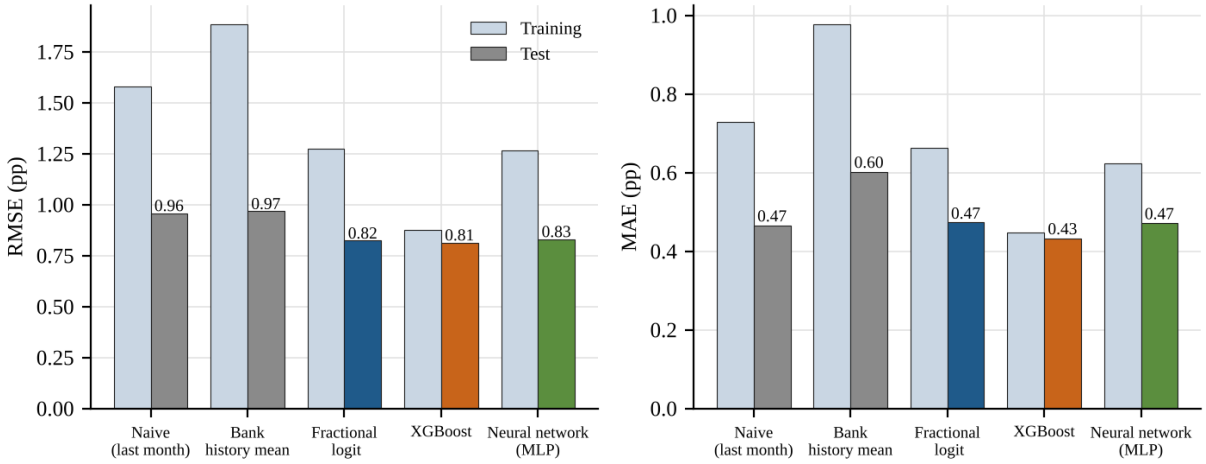
*Note.* TP, FP, FN = true positives, false positives, false negatives. Thresholds maximize $F_1$ on the validation partition (logistic regression 0.98, XGBoost 0.61, MLP 0.45). Base rate of spikes: 0.031.

## Visual Evidence

Figure 9 shows the error comparison behind Table 10: light bars are training error and colored bars test error. XGBoost has the smallest gap between them, so its regression gain is not an artifact of fitting the training months. Figure 10 gives the ROC and precision–recall curves for the spike classifiers. The XGBoost curve lies above the diagonal and the base-rate line over most of its range, the logistic curve is similar at low false-alarm rates, and the network's curve sits at or below chance, which agrees with its zero recall. Figure 11 shows the confusion matrices at the validated thresholds: the naive rule and XGBoost each find 4 of 20 spikes, but XGBoost raises 25 false alarms against 16.

## Figure 9

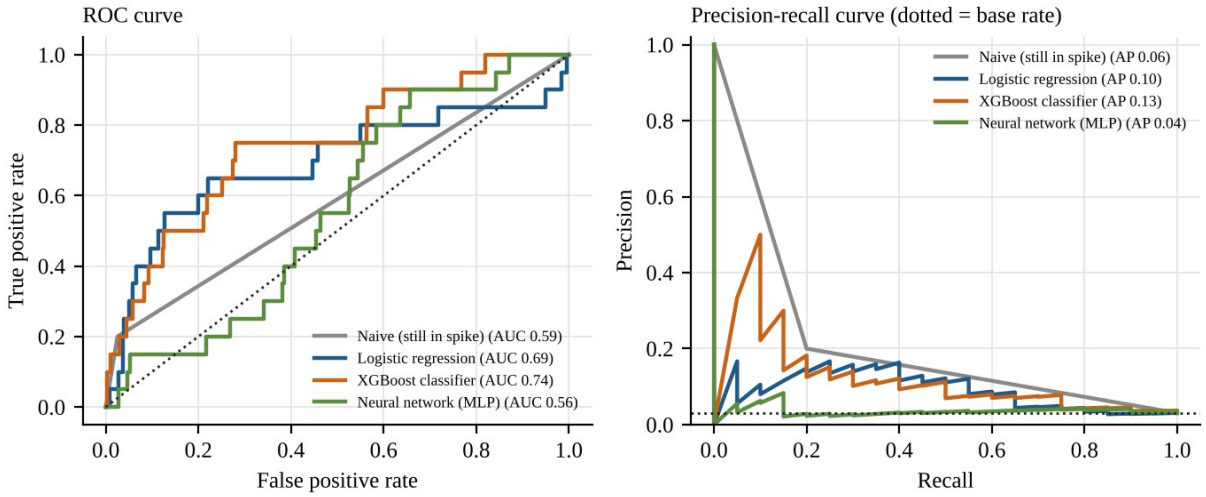
*Training and Test Error of Five Next-Month TD Predictors*




*Note.* Light bars: training (772 pairs); colored bars: rolling-origin test (653 pairs). Lower is better. Source: author's calculation from NPCI (n.d.).

## Figure 10

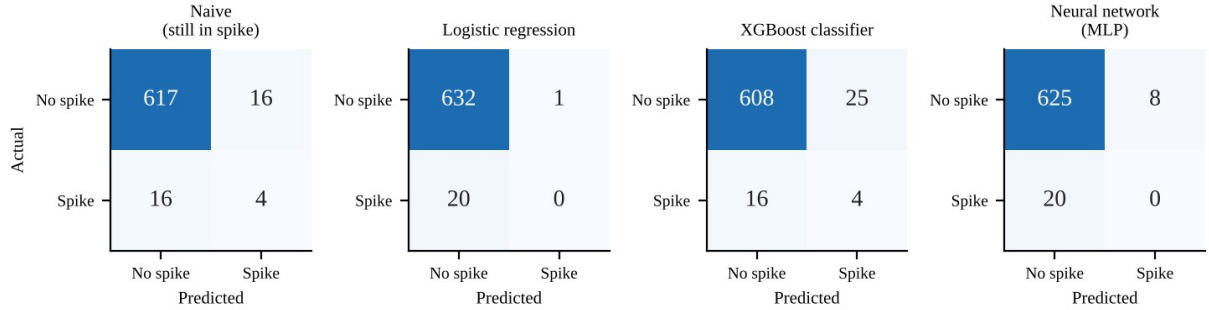
*ROC and Precision–Recall Curves for Next-Month Spike Classifiers*




*Note.* Test partition, 20 spikes in 653 pairs. The dotted diagonal marks chance and the dotted horizontal line the spike base rate (0.031). Source: author's calculation from NPCI (n.d.).

**Figure 11**

*Confusion Matrices of the Spike Classifiers at Validated Thresholds*

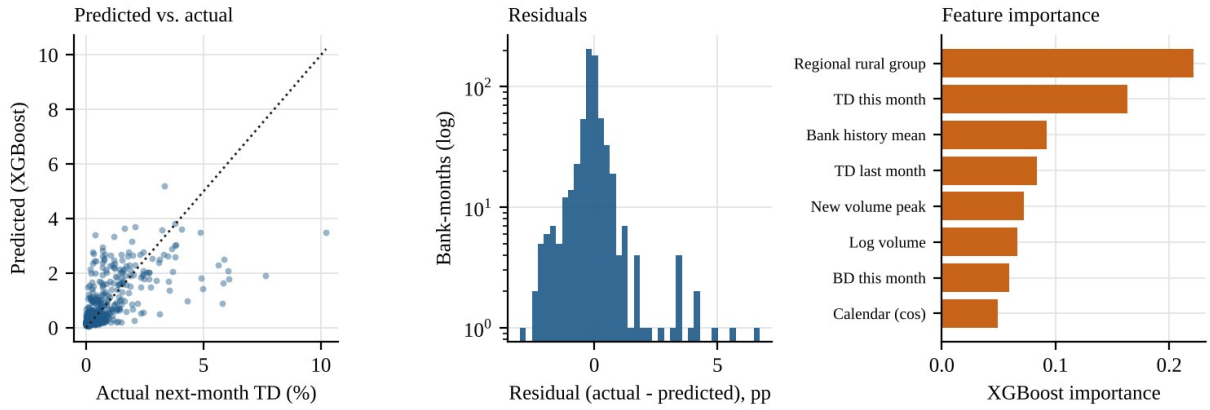



*Note.* Rows are actual and columns predicted status for the 653 test pairs. Thresholds were chosen on the validation partition. Source: author's calculation from NPCI (n.d.).

Figure 12 diagnoses the best regression model. XGBoost tracks the bulk of bank-months near zero but under-predicts the largest values: a few extreme months, up to 10% TD, are predicted at 2% to 4%. The residuals center on zero with a long right tail, and the three largest errors make up 24% of the squared error (18% for the naive rule). Without them the RMSE is 0.713 for XGBoost and 0.867 for the naive rule, so the advantage holds in ordinary months. The importance panel ranks regional-rural group membership first (0.22), then current TD (0.16) and the bank's historical mean (0.09); volume growth ranks low (0.04), consistent with RQ1.

**Figure 12**

*Diagnostics of the Best Regression Model (XGBoost)*




*Note.* Left: predicted against actual next-month TD, test partition. Center: residuals (log count axis). Right: gain-based feature importance from a model fit on all pairs. Source: author's calculation from NPCI (n.d.).

## Results by Research Question

### *RQ1: Volume Shifts and TD*

Table 12 reports fractional logit models with bank and month effects on 1,489 bank-month pairs from 56 banks. A 10% volume jump is associated with a +0.029 pp change in TD (95% CI −0.08 to +0.14) and a new peak with +0.054 pp (95% CI −0.15 to +0.26). Neither is distinguishable from zero ($p$ = .59 and .60), and the bank bootstrap agrees ($p$ = .68 and .71; AME intervals −0.10 to +0.13 and −0.15 to +0.24 pp), so $H_0$1a and $H_0$1b are not rejected at 5%. Adding last month's TD (Model B) leaves both intervals around zero ($p$ = .17 and .099), and the lag is large (coefficient 10.3, $p < .001$). This is a wide null, not proof of no effect: the power analysis showed that growth-based estimates can detect only large effects.

**Table 12**

*Fractional Logit Estimates for RQ1 (Bank and Month Fixed Effects)*

| Model and variable | Coefficient (logit) | Cluster SE | p | AME (pp) | 95% CI for AME (cluster / bootstrap) |
|---|---|---|---|---|---|
| A: ΔlnV (per 10% jump) | 0.355 | 0.662 | .59 | +0.029 | [−0.08, 0.14] / [−0.10, 0.13] |
| A: NewPeak | 0.062 | 0.118 | .60 | +0.054 | [−0.15, 0.26] / [−0.15, 0.24] |
| B: ΔlnV (per 10% jump) | 1.165 | 0.854 | .17 | +0.093 | [−0.04, 0.23] / not run |
| B: NewPeak | 0.200 | 0.121 | .099 | +0.167 | [−0.03, 0.37] / not run |
| B: $TD_{t-1}$ (share) | 10.26 | 1.49 | < .001 | — | — |

*Note.* Dependent variable: TD share. Model A n = 1,489 (56 banks); Model B n = 1,433 (55 banks). AME = average marginal effect in percentage points. Bootstrap: 400 bank-level resamples, percentile intervals.

### *RQ2: Timing of the API Controls*

Table 13 and Figure 13 show that the August 2025 estimate depends on the specification. Without a trend (A and D), TD is lower after August 2025 (−0.205 pp, 95% CI −0.361 to −0.049), but that step ranks only 20th of 21 candidate breaks, because earlier breaks give larger drops while TD was already falling. With a linear trend (B and E) the sign reverses (+0.173 pp), because TD sits above the extrapolated trend, and August ranks 3rd behind July 2025 and July 2024. The 12-month window (C) gives −0.116 pp. At 5%, every specification formally rejects $H_0$2, yet the signs disagree (three negative, two positive), and a placebo test gives the trend-

adjusted step a permutation *p* of 3/21 = .14. The pre-set rule (same sign in all five and a top-two rank) fails on both counts, so the estimates are not read as an effect of the controls. As a placebo, BD shows no level shift with a trend (−0.93 pp, CI −2.70 to +0.84) but rises 2.4 pp without one, because BD has climbed steadily, so it is a valid placebo only in trend-adjusted form. The RQ3 model, which also controls for trend, finds the post-August-2025 level above trend (logit +0.58, bootstrap interval 0.06 to 1.31), matching the trend-adjusted result: TD stopped falling around mid-2025.

**Table 13**

*Specification Check for the August 2025 Level Change in TD*

| Specification | Estimate | 95% CI | p | Rank of August (of 21) | Sign |
|---|---|---|---|---|---|
| A. Step only | −0.205 pp | [−0.361, −0.049] | .012 | 20 | − |
| B. Linear trend + step (planned) | +0.173 pp | [0.045, 0.301] | .010 | 3 | + |
| C. 12 months before vs. after | −0.116 pp | [−0.222, −0.011] | .032 | n/a | − |
| D. Log TD, step only | −37.7% | [−54.9%, −13.7%] | .006 | 19 | − |
| E. Log TD, trend + step | +30.2% | [2.8%, 64.8%] | .030 | 3 | + |

*Note.* Volume-weighted monthly TD, 32 months; Newey–West standard errors (2 lags) with *t* intervals. Rank is by absolute size among 21 candidate break months (1 = largest). Decision rule met: no.

**Figure 13**

*Estimated Level Change in TD by Assumed Break Month (Linear Trend Plus Step)*




*Note.* Each point refits the planned model with the break at that month; the circle marks August 2025. Source: author's calculation from NPCI (n.d.).

### *RQ3: Calendar Peaks*

In a fractional logit with bank effects, a linear trend, and the post indicator, the peak-month coefficient is −0.025 (SE = 0.056, $p = .66$), an AME of −0.023 pp (95% CI −0.12 to +0.08) on 342 peak bank-months. The interaction of Peak with volume exposure is +0.040 (SE = 0.061, $p = .52$), and the bootstrap gives similar $p$ values (.64 and .59). Dropping March and April 2025, near the April outage, gives −0.063 (SE = 0.065). Neither $H_0$3a nor $H_0$3b is rejected: the March and festive peaks do not raise TD once the downward trend is controlled, which agrees with Figure 4.

### *RQ4: Persistence and Heterogeneity*

The ICC of the logit of TD is 0.557 (95% CI 0.46 to 0.65), so more than half of the variation in TD belongs to the bank, and $H_0$4a is rejected. The within-bank AR(1) coefficient is 0.318 (clustered SE = 0.037, $p < .001$), a half-life of 0.60 months, so $H_0$4b is rejected: a deviation fades within about a month. The 104 spike-months in 37 banks form 69 episodes; all recovered, with a median duration of 1 month (mean 1.51, longest 4) and 68% lasting a single month. Figure 14 shows that 32% of spikes were still ongoing after one month and 6% after three. A Cox model with log volume, a peak-month start, and public-sector status gives recovery hazard ratios of 1.07 ($p = .68$), 1.27 ($p = .41$), and 0.63 ($p = .19$). None differs from 1, so $H_0$4c is not rejected on 69 episodes (Table 14 collects the RQ3 and RQ4 results).

**Table 14**

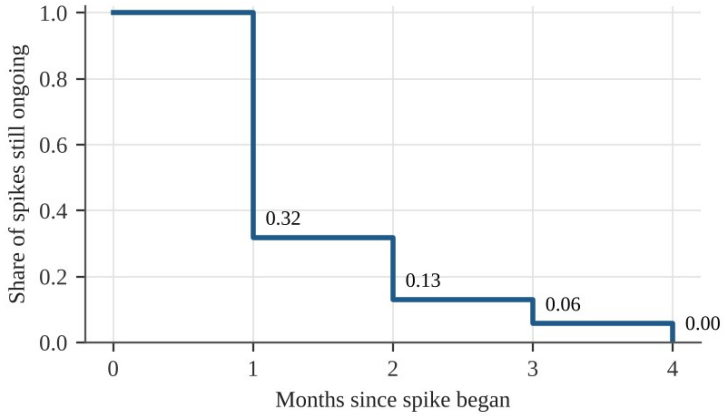
*Estimates for RQ3 and RQ4 and Hypothesis Decisions at the 5% Level*

| Question | Estimate | Uncertainty | Decision |
|---|---|---|---|
| RQ3a: Peak | β = −0.025; AME −0.023 pp | SE 0.056; $p = .66$; bootstrap $p = .64$ | Do not reject $H_0$3a |
| RQ3b: Peak × exposure | β = +0.040 | SE 0.061; $p = .52$; bootstrap $p = .59$ | Do not reject $H_0$3b |
| RQ4a: ICC | 0.557 | 95% CI [0.46, 0.65] | Reject $H_0$4a |
| RQ4b: AR(1) ρ | 0.318; half-life 0.60 months | Clustered SE 0.037; $p < .001$ | Reject $H_0$4b |
| RQ4c: Cox hazard ratios | Volume 1.07; peak start 1.27; public 0.63 | $p = .68, .41, .19$; 69 episodes | Do not reject $H_0$4c |

*Note.* ICC = intraclass correlation; AR(1) = first-order autoregression. Bootstrap: 400 bank-level resamples.

**Figure 14**

*Kaplan–Meier Curve of Spike Duration*




*Note.* The vertical axis is the share of the 69 spike episodes still ongoing; labels give the estimate at each month. Source: author's calculation from NPCI (n.d.).

## Overall Interpretation

Taken together, the results answer the four questions as follows. For RQ1, average volume shifts show no detectable effect on TD, and the data can detect only large effects, so the question remains open. For RQ2, TD fell and then flattened around mid-2025, but August 2025 cannot be told apart from nearby breaks, and the sign depends on whether a trend is assumed, so the API controls cannot be credited or blamed from these data. For RQ3, March and festive months do not raise TD once the trend is controlled. For RQ4, TD is strongly bank-specific and spikes are short. In prediction, the most important factors are persistence and bank identity: the best models win by using a bank's own recent TD and group, not by detecting volume surges, and the deep network adds nothing over a boosted-tree or logit model, in line with the benchmark literature (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2021). Performance is limited by the rarity of spikes (20 in the test months) and by the lack of daily data.

## Practical Significance

Three decisions follow. First, because TD is bank-specific and short-lived, NPCI and banks should target individual banks, mainly small regional rural and payments banks, and treat a one-month spike as a monitoring event, not a capacity trigger. Second, the boosted-tree model

cuts forecast error by about 15% against a last-month rule, enough to rank banks for review but not to raise alarms alone. Third, the data give no support for claiming that the August 2025 controls changed TD, so NPCI should evaluate any further API control with an untreated comparison, such as beneficiary-side data, before enforcing it more widely.

## Implementation and User Benefit

### Deployment Approach

The proposed deployment is a monthly batch job, because NPCI publishes data monthly and a streaming service would add cost with no new information. The job downloads or receives the new table, rebuilds the SQLite database, reruns the checks in Table 8, refits the regression and classification models on all earlier months, and writes a ranked watch list (the repository already writes *outputs/watchlist_next_month.csv*) and the monitoring figures. A dashboard built with an open-source tool such as Streamlit would display the list, bank trends, and the model's record against realized TD. The job runs on one small CPU machine; no GPU is needed.

An optional agentic layer, not part of the evaluated results, could sit on top of the job. In line with tool-using agent designs (Wang et al., 2024; Yao et al., 2023), a language-model agent would call fixed tools (read the watch list, fetch a bank's trend, compare with its group) and draft a short note for an analyst. Guardrails: every number must come from a tool call, calls are logged, and a human approves any note.

### System Integration

At NPCI, internal monthly or daily member statistics would replace the public tables, and the watch list would feed the existing capacity-review and incident process. In a bank setting, the same code can run on the bank's own switch logs, with the bank's TD compared against the group medians that NPCI publishes. Integration needs only a table with month, bank, volume, and the three decline shares. Model error is tracked each month against realized TD, and a drop in accuracy triggers a review, since NPCI could change its TD definition.

**User Interaction**

The input is the monthly table, uploaded or fetched. The output has three parts: a ranked list of banks with last month's TD, next month's predicted TD, and the spike probability; a bank page with its TD trend, group comparison, and spike history; and an optional drafted note. Users can filter by bank group and export the list. A flag appears only when the spike probability passes the validated threshold, and the page shows the false-alarm record so users can judge each alert.

**Benefits to Users**

For operations teams, the list replaces a scan of about 48 banks with a short ranked review, and it shows that failures concentrate in small banks: the ten highest-ranked banks in the September 2026 list handle only 4% of volume, so a volume-weighted average would hide them. Fewer failed payments would mean fewer retries and complaints, but the study does not estimate rupee savings. For strategy, NPCI gets a repeatable way to judge whether a control worked, with a placebo test that guards against crediting a change already under way.

**Example Use Case**

Consider an NPCI analyst on the first working day after the September 2026 table is posted. The pipeline has just been refreshed with the August 2026 data and produced Table 15. The analyst sees that Baroda UP Gramin Bank is predicted at 3.72% TD next month against 2.44% now, and that Central Bank of India, with 478 million attempts, has the highest spike probability (0.33), although below the 0.61 alert threshold, so it is not flagged. Its August TD of 2.34% means about 11.2 million failed attempts that month. The analyst opens the bank page and sends the capacity team a note asking for a review. The analyst records the case as a watch item, and when the next table arrives the record shows whether the prediction held.

**Table 15**

*Illustrative Watch List for September 2026 (Model Fit on Data Through August 2026)*

| Bank | Group | Attempts, Aug (million) | TD, Aug (%) | Predicted TD, Sep (%) | Spike probability |
|---|---|---|---|---|---|
| Baroda UP Gramin Bank | Regional rural | 109.7 | 2.44 | 3.72 | 0.19 |
| Maharashtra Gramin Bank | Regional rural | 45.9 | 1.23 | 2.14 | 0.01 |
| Rajasthan Marudhara Gramin Bank | Regional rural | 78.5 | 1.18 | 1.72 | 0.01 |
| Central Bank of India | Public sector | 478.3 | 2.34 | 1.67 | 0.33 |
| Kerala Gramin Bank | Regional rural | 68.6 | 0.84 | 1.65 | 0.10 |
| Sarva Haryana Gramin Bank | Regional rural | 33.2 | 0.68 | 1.63 | 0.15 |

*Note.* Output of code/benchmark.py for 48 banks with complete features; ranked by predicted TD. This is an illustration of the deployed output. The September 2026 outcomes were not available when the model was fit, so these predictions are not validated.

## Limitations and Further Improvements

### Limitations

Three limitations concern the data. The data are monthly bank aggregates on the remitter side, so findings describe banks and months, not users or transactions. The top-50 list changes, so the panel is unbalanced (39 of 56 banks appear in all 32 months). TD is defined by NPCI, and a change in classification practice would look like a real change in failure rates, which public data cannot rule out. Two judgment calls remain unverified against the portal: merging one bank into another (12 of 1,555 bank-months, 0.8%) and the month label of the August 2026 file.

Four limitations concern the methods and models. The growth-based RQ1 estimates are underpowered, and RQ2 has one system and one date with no untreated group. The test partition holds only 20 spikes, so every spike metric is imprecise, and the XGBoost classifier overfits its training spikes. The neural network was tuned on a short validation window. Bank groups are coded by rule, and the bootstrap resamples whole banks and is not a wild cluster bootstrap. Alternative spike thresholds and leave-one-bank-out checks were not run for this draft, and all results are associations from observational data.

### Impact of Limitations

These limits mainly affect how far the findings travel. The null results for RQ1 and RQ3 mean “no effect large enough to detect here,” not “no effect,” and a small number of rare jumps at small banks could matter operationally while remaining invisible in an average. The RQ2 conclusion rests on a pre-set rule, and a different rule could read the same data differently. Prediction results are probably optimistic for spikes in new regimes, because the test months were quiet compared with early 2024. The results apply to NPCI’s published remitter tables and should not be extended to merchants, apps, or individual users.

### Future Improvements

The most useful improvement is more data: the beneficiary-side tables and later months would raise the power of RQ1 and allow a difference-in-differences contrast for RQ2. Further checks should include alternative spike definitions, leave-one-bank-out estimates, a wild cluster bootstrap, and a winsorized outcome. On the modeling side, calibrated spike probabilities, count-based forecasts of failed transactions, and a longer validation window with more spikes would help. The bank rename and the August 2026 label should be confirmed with NPCI.

### Future Scope

The study can be extended in three directions. Transaction-level or daily data from NPCI or a bank would allow tests of capacity at the level where outages happen and a stronger design for the API controls, for example by comparing affected and unaffected API types. The same pipeline could cover other payment systems that publish decline statistics. Finally, the proposed monitoring service and its drafting agent could be built and tested with analysts, measuring whether the watch list shortens the time to respond to a spike.

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## Appendix A: Code Excerpts

The complete code is in the repository (*code/*). The excerpts below produce Table 12 (RQ1) and Table 10 (benchmark), and are referred to in the Materials and Method section. The analysis used Python with pandas, statsmodels, scikit-learn, and XGBoost, and random seed 42.

```
# RQ1: fractional logit with bank and month fixed effects, bank-clustered SE
(final_analysis.py)
X = pd.get_dummies(df[["entity","month"]], drop_first=True, dtype=float)
X = sm.add_constant(pd.concat([df[["dlnV","newpeak"]].reset_index(drop=True),
X.reset_index(drop=True)], axis=1))
m = sm.GLM(df.p.values, X, family=sm.families.Binomial()).fit(
        cov_type="cluster", cov_kwds={"groups": pd.factorize(df.entity)[0]})
ame_10pct = m.params["dlnV"] * mean_wt * 100 * np.log(1.1)   # mean_wt = mean(mu*(1-mu))

# Benchmark: rolling-origin prediction, train strictly before each target month
(benchmark.py)
def rolling(fn, kind, hp, target_months):
    out = []
    for m in target_months:
        tr = M[M.next_month < m]; te = M[M.next_month == m]
        out.append(pd.Series(fn(kind, tr, te, hp), index=te.index))
    return pd.concat(out)
P["XGBoost"] = rolling(fit_predict_reg, "xgb", best[("reg","xgb")],
TEST).loc[T.index].values

# Scaler fit on training rows only (leakage control) for the neural network and logistic
regression
sc = StandardScaler().fit(tr[FEATS]); Xtr, Xte = sc.transform(tr[FEATS]),
sc.transform(te[FEATS])
m = MLPRegressor(hidden_layer_sizes=(16,), alpha=1e-3, learning_rate_init=.003,
max_iter=400,
                 early_stopping=True, validation_fraction=.15, n_iter_no_change=20,
random_state=SEED)
```

## Appendix B: Database Schema, Queries, and Tuning Results

Table B1 shows the SQLite schema built by *code/db_setup.py*, and Table B2 the validation scores used to choose hyperparameters.

## Table B1

*Database Tables and Example Queries*

| Object | Columns / purpose | Example query |
|---|---|---|
| panel | month, rank, bank_raw, bank, entity, is_bank, volume_mn, approved_pct, bd_pct, td_pct, debit_reversal_mn, debit_reversal_success_pct, source_file, month_source, n_member_rows; key (month, entity) | SELECT COUNT(*) FROM panel; (1,598) |
| entity_crosswalk | bank_raw (key), bank, entity, is_bank | SELECT entity, COUNT(*) FROM entity_crosswalk GROUP BY entity HAVING COUNT(*) > 1; |
| file_log | source_file (key), month, month_source, member_rows | SELECT * FROM file_log WHERE month_source <> 'sheet title'; |
| v_bank_month (view) | Bank rows only: month, entity, volume_mn, approved_pct, bd_pct, td_pct | SELECT month, SUM(volume_mn*td_pct)/SUM(volume_mn) AS td_weighted FROM v_bank_month GROUP BY month; |

## Table B2

*Validation Scores Used for Hyperparameter Selection (January–June 2025)*

| Model | Setting | Score | Selected |
|---|---|---|---|
| XGBoost regression | Depth 2, 150 trees | RMSE 1.442 | |
| XGBoost regression | Depth 2, 300 trees | RMSE 1.504 | |
| XGBoost regression | Depth 3, 150 trees | RMSE 1.425 | Yes |
| XGBoost regression | Depth 3, 300 trees | RMSE 1.465 | |
| XGBoost regression | Depth 4, 150 / 300 trees | RMSE 1.460 / 1.498 | |
| MLP regression | 16 units, penalty 0.001 / 0.1 | RMSE 1.249 / 1.249 | Yes (0.001) |
| MLP regression | 32 and 16 units, penalty 0.001 / 0.1 | RMSE 1.312 / 1.306 | |
| XGBoost classifier | Depth 3, 300 trees (others 0.129–0.194) | PR-AUC 0.198 | Yes |
| MLP classifier | 32 and 16 units, penalty 0.001 | PR-AUC 0.173 | Yes |
| Logistic regression | C = 0.01 / 0.1 / 1.0 | PR-AUC 0.215 / 0.216 / 0.217 | Yes (1.0) |

*Note.* RMSE and PR-AUC on the validation partition, 294 pairs and 10 spikes. Lower RMSE and higher PR-AUC are better. Full grid in code/benchmark_results.json.